

Newman-Penrose Formalism Calculations

Stephani et al., Equation (31.60)

Metric

$$ds^2 = \frac{2(az^2 - b)}{x^2} dt \otimes dt + \left(-\frac{2}{2e^2x^6 - cx^5 - ax^4} \right) dx \otimes dx + (-4e^2x^2 + 2cx + 2a) dy \otimes dy + \left(-\frac{2}{ax^2z^2 - bx^2} \right) dz \otimes dz$$

Coordinates

$$(x^\mu) = (t, x, y, z)$$

Null Tetrad

Covariant components

$$l = \left(\frac{\sqrt{-az^2 + b}}{x} \right) dt + \left(\frac{1}{\sqrt{-az^2 + bx}} \right) dz$$

$$n = \left(\frac{\sqrt{-az^2 + b}}{x} \right) dt + \left(-\frac{1}{\sqrt{-az^2 + bx}} \right) dz$$

$$m = \left(\frac{1}{\sqrt{-(2e^2x^2 - cx - a)x^2}} \right) dx + \left(\frac{i\sqrt{-(2e^2x^2 - cx - a)x^2}}{x} \right) dy$$

$$\bar{m} = \left(\frac{1}{\sqrt{-(2e^2x^2 - cx - a)x^2}} \right) dx + \left(-\frac{i\sqrt{-(2e^2x^2 - cx - a)x^2}}{x} \right) dy$$

Contravariant components

$$l = \left(\frac{\sqrt{-az^2 + bx}}{2(az^2 - b)} \right) dt + \left(-\frac{(az^2 - b)x}{2\sqrt{-az^2 + b}} \right) dz$$

$$n = \left(\frac{\sqrt{-az^2 + bx}}{2(az^2 - b)} \right) dt + \left(\frac{(az^2 - b)x}{2\sqrt{-az^2 + b}} \right) dz$$

$$m = \left(-\frac{1}{2} \sqrt{2e^2x^2 - cx - ax^2} \right) dx + \left(-\frac{i}{2\sqrt{2e^2x^2 - cx - a}} \right) dy$$

$$\bar{m} = \left(-\frac{1}{2} \sqrt{2e^2x^2 - cx - ax^2} \right) dx + \left(\frac{i}{2\sqrt{2e^2x^2 - cx - a}} \right) dy$$

Directional Derivatives

$$D = \frac{\sqrt{-az^2 + bx}}{2(az^2 - b)} \partial_t - \frac{(az^2 - b)x}{2\sqrt{-az^2 + b}} \partial_z$$

$$\Delta = \frac{\sqrt{-az^2 + bx}}{2(az^2 - b)} \partial_t + \frac{(az^2 - b)x}{2\sqrt{-az^2 + b}} \partial_z$$

$$\delta = -\frac{1}{2} \sqrt{2e^2x^2 - cx - ax^2} \partial_x - \frac{i}{2\sqrt{2e^2x^2 - cx - a}} \partial_y$$

$$\bar{\delta} = -\frac{1}{2} \sqrt{2e^2x^2 - cx - ax^2} \partial_x + \frac{i}{2\sqrt{2e^2x^2 - cx - a}} \partial_y$$

Spin Coefficients

$$\rho = 0$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = 0$$

$$\lambda = 0$$

$$\nu = 0$$

$$\alpha = -\frac{(12\,e^2x^2 - 5\,cx - 4\,a)x}{8\sqrt{2\,e^2x^2 - cx - a}}$$

$$\beta = \frac{(12\,e^2x^2 - 5\,cx - 4\,a)x}{8\sqrt{2\,e^2x^2 - cx - a}}$$

$$\tau = -\frac{1}{2}\sqrt{2\,e^2x^2 - cx - ax}$$

$$\pi = \frac{1}{2}\sqrt{2\,e^2x^2 - cx - ax}$$

$$\epsilon = -\frac{axz}{4\sqrt{-az^2 + b}}$$

$$\gamma = -\frac{axz}{4\sqrt{-az^2 + b}}$$

Ricci Tensor Components

$$\Phi_{00} = 0$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = 0$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = 0$$

$$\Lambda = 0$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\Psi_2 = \frac{1}{4} (4 e^2 x - c) x^3$$

$$\Psi_3 = 0$$

$$\Psi_4 = 0$$

Newman-Penrose Equations

$$NP6 = \frac{1}{2} (e^2 x^2 + a) x^2$$

$$NP7 = -\frac{1}{4} (8 e^2 x^2 - 3 c x - 2 a) x^2$$

$$NP8 = \frac{1}{4} (8 e^2 x^2 - 3 c x - 2 a) x^2$$

$$NP12 = \frac{(8 e^4 x^4 - 8 c e^2 x^3 - 4 a e^2 x^2 + 3 c^2 x^2 + 6 a c x + 4 a^2) x^2}{8 (2 e^2 x^2 - c x - a)}$$

$$NP16 = -\frac{1}{4} (8 e^2 x^2 - 3 c x - 2 a) x^2$$

$$NP17 = -\frac{1}{4} (8 e^2 x^2 - 3 c x - 2 a) x^2$$

Bianchi Identities

$$BI7 = -\frac{1}{2} \sqrt{2 e^2 x^2 - c x - a} e^2 x^5$$

$$BI8 = -\frac{1}{2} \sqrt{2 e^2 x^2 - c x - a} e^2 x^5$$

Petrov Type

Type "D"